

The USAPhO Exam

What it is, how it is run, and how to write answers that earn points

For official information, use the AAPT (American Association of Physics Teachers) website. Trust that website above anything in this handout.

www.aapt.org/physicsteam

The pages worth bookmarking are [Rules](#), [For Students](#), and [Past Exams](#). Everything below comes from those pages or from the instruction pages of recent exams, but the exam changes — check before you plan around any of it.

1 The Short Version

The USAPhO is a three-hour, six-problem, free-response exam worth 150 points. It runs in two 90-minute halves of three problems each, with a break between, and you may not return to the first half. You qualify by clearing the cutoff on the $F = ma$ exam in February; the USAPhO is in April. It is proctored in person. The questions are delivered on a computer, but **you write your solutions by hand on blank paper the proctor gives you**, and those pages are scanned and graded by humans. **You get a reference sheet** with constants and a few approximations. A basic scientific calculator is allowed. Almost nobody finishes, and partial credit is most of the game.

2 Part I: The Logistics

How do I qualify?

By scoring at or above the cutoff on the $F = ma$ exam. AAPT sets that cutoff after the exam, at whatever score invites roughly 400–500 students; recent cutoffs have been 14–18 out of 25. There is no other route in — you cannot register for the USAPhO directly. See the *F = ma Exam FAQ* handout for everything about that exam.

When is it?

April, a few weeks after $F = ma$ results come out. In 2026 it was **April 10**, inside a 1:00–5:00 pm EDT window (the exam is three hours plus a break; your proctor picks the start time inside the window). Watch AAPT’s “Important Dates” page — in 2026 the date moved once, from April 2 to April 10, to avoid a Passover conflict.

What is the shape of the exam?

	Part A	Part B
Length	90 minutes	90 minutes
Problems	3 (A1, A2, A3)	3 (B1, B2, B3)
Points each	25	25
Part total	75	75

150 points total. Each problem has several lettered subparts. The instructions say plainly that problems worth the same points “may differ in difficulty,” so the order on the page is *not* a difficulty ranking.

Part A and Part B are sealed off from each other. You may not look at Part B during Part A, and after the break you may not go back to Part A — your Part A pages have already been collected. Treat them as two separate 90-minute exams. Whatever you did not finish in Part A is gone.

What topics does it cover?

All of introductory physics: mechanics, electricity and magnetism, thermodynamics and statistical physics, waves, optics, and relativity and modern physics. Unlike $F = ma$, it is **not** mechanics-only, and unlike $F = ma$, **calculus is expected**. You will integrate, differentiate, expand in small parameters, and solve differential equations. The 2026 exam included a geometrical optics problem, a problem about motion on a cone, a blackbody radiation problem, and a rigid-body problem, among others.

Is it on paper or on a computer?

Both, and this is the part people get wrong. Since the 2026 cycle AAPT states that the $F = ma$ and USAPhO exams are “conducted online via Educational Vistas only,” with “no printable versions of the exams available.” What that means in practice, from the 2025 and 2026 student instructions:

- You sit at a designated desktop, laptop, or Chromebook. Phones, tablets, and smartwatches are handed in before the exam starts.
- **You read the problems on the screen.** When the exam begins you click the link for Part A and a 90-minute timer starts.
- **You write your solutions by hand**, on blank paper the proctor hands you, along with a printed instruction page, the reference sheet, and a cover sheet for each problem.
- At the end of each part the proctor collects your written pages and scans them. Those scans are what get graded.

The delivery is online; the answering is not. You are still writing physics on paper with a pen, and a human is still reading it.

Do I get scratch paper, and how should I use it?

Yes, from the proctor only — “Only scratch paper provided by the proctors may be used” — and it is all collected at the end. You take none of it home.

The rule that matters is the one about what gets *graded*: “All work must be done on the provided blank pages. . . Work outside of these pages will not be graded,” and when the proctor collects your solutions, “Do not include scratch work.” **Scratch paper is invisible to the grader.**

Use scratch paper for deciding, not for solving. Scratch is the right place to try three approaches and see which two collapse. Once you know which one works, **switch to the answer pages and do the real derivation there.** Do not solve the whole problem on scratch and then copy it over — that is doing the problem twice, and you do not have time to do six problems twice. Grinding through a long integral on scratch and transcribing only the final line is the worst of both worlds: you spent the time and the grader cannot see the work you spent it on.

Is there a formula sheet?

Yes — a real difference from $F = ma$, which gives you nothing. The USAPhO reference sheet is printed in the instructions and you may use it for both parts. In 2026 it held:

- **Constants:** g , G , $k = 1/4\pi\epsilon_0$, $k_m = \mu_0/4\pi$, c , k_B , N_A , R , σ , e , 1 eV in joules, h , and m_e .
- **Approximations:** $(1+x)^n \approx 1+nx + \frac{1}{2}n(n-1)x^2$, $e^x \approx 1+x+x^2/2+x^3/6$, $\sin\theta \approx \theta - \theta^3/6$, and $\cos\theta \approx 1 - \theta^2/2$.

That is the whole sheet. **It is a constants sheet, not a formula sheet.** There is no $\frac{1}{2}mv^2$, no Maxwell’s equations, no moments of inertia, no thermodynamic relations, no lens equation. Some years have also carried a short table of integrals and trigonometric sum identities; the 2026 sheet did not. Do not count on it. Know your formulas.

That they hand you those four approximations is a hint about the exam’s taste: **expansions in a small parameter show up constantly.** When a problem hands you a condition like $r \ll R$, it is telling you which expansion to run.

What are the calculator rules?

A handheld calculator is allowed, but it “may not be a graphical calculator, its display has no more than three lines,” and its memory must be cleared of data and programs before the exam. No symbolic math, no programming, no graphing. Calculators may not be shared. Bring your own; the proctor will not have one. You will use it much less than on $F = ma$ — most answers are symbolic.

What else is banned?

Books, tables, collections of formulas, notes — anything beyond the provided reference sheet. Phones and other internet-connected devices are handed in and may not be used at any time while exam materials are present. You must stay in the exam room for the entire testing period.

How is it proctored?

In person and live, like $F = ma$. Proctors need at least a two-year degree, need not be physics teachers, and **may not be a parent, relative, or close friend** of a student taking the exam. For-profit test prep centers are not allowed to proctor. If you qualify, AAPT emails you and your proctor, and usually the same teacher who ran your $F = ma$ administration runs this one.

Can I talk about the problems afterward?

Not until the embargo lifts, which is typically the day after the exam — the 2025 exam told students not to discuss any part of it until April 11, the day after. Violations can mean disqualification. Solutions are posted publicly a few weeks later.

What happens after?

Results and an honors list come out in May. About 20 students, chosen on combined USAPhO and $F = ma$ performance, are named to the **U.S. Physics Team** and invited to a ten-day training camp at the University of Maryland in late May. AAPT's standing description of what happens next is that “at the end of the camp, five students will be selected to represent the team at the International Physics Olympiad” — those five come out of the exams given during the camp itself.

2026 did it differently, and it is not yet clear whether that sticks. A new exam, the **USAPhO Plus**, was given on May 10 to the 20 students *already* named to the team, and it — not the camp — chose the five travelers and one alternate. The camp, renamed a Training & Development Camp, ran afterward (May 30–June 9) to prepare that delegation and to develop future competitors. The 2026 team also competed at the **European Physics Olympiad** in Gothenburg rather than the IPhO.

There was no USAPhO Plus in 2024 or 2025; in 2025 the five IPhO travelers were named after camp in the usual way. **None of this changes anything you do on exam day** — but if you are in range of the top 20, read AAPT's news page for your year rather than assuming either pattern.

One feature of the 2026 USAPhO Plus is worth knowing about regardless: alongside the usual theory problems it included a problem with an **experimental component** — analyzing experimental-style data, constructing and interpreting plots, extracting physical parameters, and evaluating uncertainties. Data analysis is also listed in the scope of the USAPhO itself, and it is the part of the syllabus American students most often skip.

3 Part II: How to Write an Answer

Start by picturing the grader

Your solution is scanned and read by a person who has already read a few hundred versions of the same problem, working from a rubric that assigns points to specific milestones: the right principle, the right setup, the right intermediate expression, the right final answer. They are not trying to reconstruct your thinking. They are trying to find the milestones.

Everything below follows from that one fact.

The five habits.

1. **Draw a labeled diagram and define your symbols on it.** Every angle, distance, and force you use should be visible somewhere. A grader who cannot tell what your ℓ is cannot give you credit for an equation containing ℓ .

2. **Name the principle before you use it.** One short line — “torque balance about the contact point in the rotating frame,” “energy conservation between release and the bottom,” “Gauss’s law on a cylinder of radius r .” This is usually worth points on its own, and it costs eight seconds.
3. **Work symbolically; substitute numbers last.** Symbols show the grader the structure. A page of decimals shows them nothing, and one arithmetic slip poisons everything downstream.
4. **Box the final answer of every subpart.** The instructions explicitly ask for this: “To help with grading, please draw a clear box around your final answer to each subpart.” An unboxed answer buried mid-page gets missed.
5. **One problem per sheet, header on every page.** Never mix two problems on one sheet. Label every page in the upper right.

What the page looks like

The exact answer-sheet format has drifted — for years AAPT printed framed answer sheets with an information block, and more recently you get a cover sheet per problem plus blank paper. What has not changed is the header AAPT asks for in the upper right corner of every page:

Student AAPT ID #	Proctor AAPT ID #
A1 — 1/3	

That last line is the one people forget and the one that matters most: **problem number, then page n of N for that problem**. The scans get split apart and sorted by problem before grading. A page labeled only “2” can end up in the wrong pile, and a page in the wrong pile scores zero.

Practical rules that follow from the pages being scanned and sorted:

- **Write on one side only.** Older instructions said so outright; graders scan fronts.
- **Number every page as n/N and fill in N at the end**, when you know it.
- **Press hard and write dark.** Faint pencil scans badly. Pen is safer, which is why crossing out beats erasing.
- **Leave the margins alone.** Anything in the outer half inch may be cropped.
- If you continue a problem on extra pages, **say so at the bottom of the previous page** (“continued on A1 – 3/3”).

What helps your score, and what costs you

Nothing here is an absolute rule. Graders are physicists reading in good faith: if your R is obviously the radius in your figure, nobody is going to pretend otherwise. But every item in the right-hand

column makes them work harder to find the points you earned, and across a stack of three hundred papers that is a cost somebody pays.

Helps you	Costs you
A named principle, even with no algebra after it	A page of algebra with no statement of what is being computed
A correct setup with the integral left unevaluated	A setup you did in your head, so only the result of it is on the page
A labeled diagram defining your variables	Symbols you never define, leaving the reader to guess which length is which
“I could not do (a); assume the answer is X ” and then a correct (b)	A blank (b) because (a) did not work out
One clearly boxed answer per subpart	Three candidate answers, none crossed out
A wrong answer whose method is visible and mostly right	A wrong answer whose method is invisible

Never leave two live answers on the page. If you tried something, abandoned it, and tried again, put one clean line through the dead attempt. Graders are generally instructed to grade what you present, and if you present two contradictory answers they are entitled to grade the wrong one. Crossing out is free. Do it.

Getting past a part you cannot do

This is the single most valuable habit on this exam, and it is the one students most often fail to use. Subparts are usually graded independently. If part (b) says “using your result from part (a),” you do not need part (a) to be *right* — you need it to be *stated*. So:

When you are stuck on (a) and (b) depends on it, write:

“Could not complete (a). For (b) I assume the result of (a) is $T = \dots$ ”

and then do (b) properly. Full marks on (b) are still available. Two more versions of the same move:

- **Carry a symbol you cannot evaluate.** If you cannot find the moment of inertia, call it I , solve the rest of the problem in terms of I , and box that. You will lose the points for I and keep the rest.
- **Do the special case.** If the general problem is beyond you, solve it for the symmetric case, or to leading order in the small parameter, and say clearly that this is what you did. It is worth far more than a blank.

Two more things that reliably earn points

Check limiting cases, out loud, on the page. When you get an answer, test it: does it reduce to the earlier part when the correction goes to zero? Is the sign right? Do the units work? Writing “as $h \rightarrow 0$ this reduces to part (a) ✓” does two jobs: it catches your own errors, and it shows a grader who is deciding between two rubric levels that you understand what you produced.

Say what you are about to do before you do it. Two-word headers — “Setup,” “Torque about P ,” “Evaluate,” “Check” — turn a wall of algebra into a readable argument. On a page that gets four minutes of attention, structure is worth points.

4 Part III: Strategy for the Three Hours

The first five minutes: orient, do not solve

Do not start problem A1 the moment the clock starts. Spend the first three to five minutes reading all three problems in the part. You are looking for:

- Which topic each problem is (mechanics? thermo? circuits?).
- How many subparts each has, and whether the early subparts look approachable.
- Which problem is *yours* — the one in your strongest area.

This costs five minutes out of ninety and it protects you from the worst outcome on this exam: spending forty minutes on the hardest problem of the set because it happened to be printed first, and never reading a problem you could have solved.

The order on the page is not the order of difficulty. AAPT says so explicitly. A3 is often easier than A2.

Problem by problem, not skipping around

Once you have chosen an order, **work one problem at a time, start to finish.** Long free-response physics is not like multiple choice: there is a large cost to reloading a problem’s setup into your head, so ping-ponging between problems wastes several minutes each time.

Within a problem, though, skip freely. If part (b) is stuck, read (c) and (d) immediately — they are often independent, or they depend on (b) in a way you can bridge with an assumed result. Some of the cheapest points on the exam sit in the last subpart of a problem whose middle nobody finished.

A time budget

Ninety minutes, three problems: **30 minutes each**, and the discipline is in the checkpoints, not the plan.

Clock	Where you should be
0:00–0:05	Read all three problems. Pick an order.
0:05–0:32	First problem (your best topic).
0:32–0:59	Second problem.
0:59–1:26	Third problem.
1:26–1:30	Harvest: box unboxed answers, fix headers, cross out dead work.

Hard stop at 30 minutes per problem, even mid-derivation. Write down where you got to, box whatever partial result you have, and move on. If you have time at the end you can come back — and you will come back with a fresh head, which is often what you needed. The alternative, sinking 50 minutes into one problem, costs you the first easy subparts of a problem you never read.

The five-minute rule for a stuck subpart: if five minutes of real effort produces no new equation, that subpart is over. Write what you know, state your assumption, and go to the next subpart.

The last four minutes of each part

Stop doing physics and go harvest points:

1. Box every final answer that is not boxed.
2. Put the problem number and n/N on every page; fill in N .
3. Cross out every abandoned attempt so nothing contradictory is left live.
4. For every subpart you left blank, write *one line* — the principle you would have used, the diagram you would have drawn, the form you expect the answer to take. A sentence beats a blank, and sometimes it is a rubric line.
5. Sort your pages by problem, then by page number.

Calibrate: you are not going to get 150

This matters, because the biggest self-inflicted wound on this exam is a student who treats an unfinished problem as a failure and spirals. The USAPhO is built so that the top students in the country do not finish it.

Solving two problems completely and making real progress on two more is an excellent performance. A blank subpart is normal. Three blank subparts is normal. **Your job is not to finish the exam — it is to collect the most points available to you in 180 minutes.** Those are different goals and they lead to different behavior: the second one makes you leave a hard problem, read the next one, and pick up its first two subparts.

5 Part IV: What Score Do I Need?

AAPT does not publish USAPhO score cutoffs, and has said it deliberately does not release numerical scores or detailed breakdowns. What it does publish is the *structure* of the awards, which is enough to aim by. Awards are given by percentile of the students who take the exam:

Award	Share of takers	2026: about
Gold Medal	top 10–12%	top 38 students
Silver Medal	next 14–16%	next 48
Bronze Medal	next 20–22%	next 63
Honorable Mention	next 24–26%	next 79
Distinction	remainder	the rest

Above all of that sits the **U.S. Physics Team**: about 20 students invited to camp, on combined USAPhO and F = ma performance.

2026 numbers: 440 students qualified on F = ma and 317 actually sat the USAPhO.

Two things fall out of that table. First, **a medal of some color goes to roughly the top 47% of the field**, and some honor goes to nearly three quarters of it. Second, **the field is 300-odd people who already finished in the top 7% of F = ma takers**. Middle of this pack is a genuinely strong physics student.

What those percentiles have meant in points

The table below is built from the actual USAPhO score files for 2014–2018. AAPT has never published cutoffs, so these are reconstructed the way AAPT sets them in the first place: take that year’s real distribution of scores and cut it at the published percentile bands. Through 2017 the exam was out of **200** points (four Part A problems at 25 and two Part B problems at 50); since 2018 it has been out of **150**. The last block converts everything to today’s 150-point scale so the years can be compared.

Year	Took it	Out of	Median	Score needed for			
				HM	Bronze	Silver	Gold
2014	368	200	64	50	68	89	118
2015	363	200	67	47	71	95	129
2016	326	200	92	69	94	116	142
2017	418	200	49	32	52	71	89
2018	382	150	40	25	42	60	84
<i>On today’s 150-point scale:</i>				25–52	39–70	53–87	67–106
<i>Typical year:</i>				35	50	67	88

Three things to take from that table.

The cutoffs move enormously. Gold ranged from 67 to 106 on today’s scale — 2016 was a gentle paper (median 46% of the points) and 2017 was brutal (median 25%). There is no fixed number to chase. **Aim at a fraction of the paper, not a score.**

Everything is lower than you would guess. In four of these five years the median was between a quarter and a third of the available points, and gold sat near 55–60%. Even the top paper in 2018 scored 124 out of 150.

In problems, on the 150-point scale: honorable mention is about a problem and a half's worth of credit, bronze about two, silver about two and a half, and gold about three and a half out of six. Note that none of those is a whole number of clean solutions — they are all reached by finishing some problems and collecting partial credit on others.

Where the points actually were in 2018. Mean score on each 25-point problem, and the share of the field that scored a flat zero on it:

	A1	A2	A3	B1	B2	B3
Mean score (of 25)	11.9	6.4	3.2	9.5	5.0	8.9
Scored zero	27%	10%	45%	16%	52%	15%

Half the field scored nothing at all on A3 and on B2. Look at the order, too: A3 was by far the hardest problem in Part A, and B3 was worth nearly twice B2. A student who worked strictly in order, and stayed on A3 until it broke, walked past the two problems in Part A where the points were. **Scattered partial credit across all six problems outscores heroic effort on one.**

6 Part V: What a Real Submission Looks Like

Everything above is abstract until you see a page. So here is one problem worked the way you would actually work it under time pressure — including a wrong turn — followed by how it would be graded.

The problem is **2014 USAPhO, Problem A1**, which is a good one to practice on: two subparts, the second a genuine correction to the first, and a trap in the middle that most people fall into.

Question A1 (2014 USAPhO). A unicyclist of total height h goes around a circular track of radius R while leaning inward at an angle θ to the vertical. The acceleration due to gravity is g .

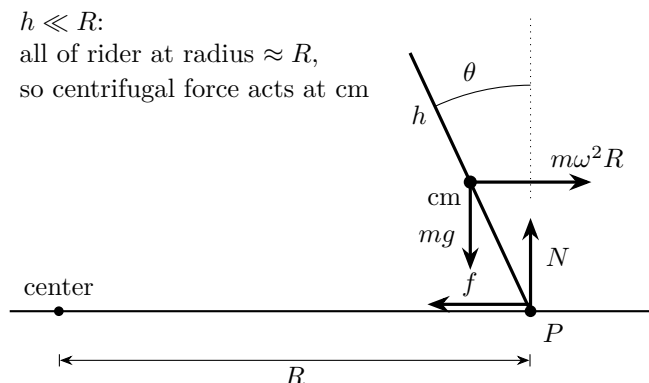
- Suppose $h \ll R$. What angular velocity ω must the unicyclist sustain?
- Now model the unicyclist as a uniform rod of length h , where h is less than R but not negligible. This refined model introduces a correction to the previous result. What is the new expression for the angular velocity ω ? Assume that the rod remains in the plane formed by the vertical and radial directions, and that R is measured from the center of the circle to the point of contact at the ground.

The submission

What follows is typeset so you can read it, but the content, the length, and the mess are what a good submission actually looks like. The notes in red are mine, pointing out the choices that are earning points; they would not be on your page.

Student AAPT ID # Proctor AAPT ID #
A1 — 1/2

(a) Rotating frame, rider static. Radial-vertical plane:



↑ The diagram is the setup. Every symbol used below — P , θ , cm, R — is defined on it, and the choice of frame is stated in the first line.

Let cm be a distance ℓ up the rod from P . Torque about P (rider is static in this frame, so $\tau_{\text{net}} = 0$; N and f act at P and contribute nothing):

$$\tau = m\omega^2 R (\ell \cos \theta) - mg (\ell \sin \theta) = 0$$

↑ “Torque about P ” named, and the reason N and f drop out given in half a line. This is a rubric line.

$$\omega^2 = \frac{g \tan \theta}{R}$$

~~$\omega^2 = g \cot \theta / R$~~ — lever arms swapped ← First attempt, caught and crossed out. See the check below.

$$\boxed{\omega = \sqrt{\frac{g \tan \theta}{R}}}$$

Check: $\theta \rightarrow 0$ (upright) $\Rightarrow \omega \rightarrow 0$. ✓ $\theta \rightarrow 90^\circ \Rightarrow \omega \rightarrow \infty$. ✓ Units: $\sqrt{(\text{m/s}^2)/\text{m}} = \text{s}^{-1}$. ✓

↑ This check is what caught the swapped lever arms. Three lines, thirty seconds, and it is the difference between 8/8 and 3/8 on this part.

Student AAPT ID # Proctor AAPT ID #
A1 — 2/2

(b) Now $h \not\ll R$, so different parts of the rod are at different radii and the centrifugal force is *not* uniform.

Idea: ~~cm sits at radius $R - \frac{h}{2} \sin \theta$, so replace $R \rightarrow R - \frac{h}{2} \sin \theta$ in (a).~~

← The trap, tried and abandoned. Note the one-line reason written next to it rather than a silent erasure:

No — torque involves the product rz , and $\langle rz \rangle \neq \langle r \rangle \langle z \rangle$. Must integrate.

Setup. Mass element a distance s along the rod from P :

$$r = R - s \sin \theta, \quad z = s \cos \theta, \quad dm = \frac{m}{h} ds, \quad 0 \leq s \leq h$$

↑ Each of the three is a rubric line on its own. Even if the integral had gone nowhere, these are already worth points.

Centrifugal torque about P . Force $\omega^2 r dm$ outward, lever arm z :

$$\begin{aligned} \tau_c &= \int_0^h \omega^2 (R - s \sin \theta) (s \cos \theta) \frac{m}{h} ds \\ &= \frac{m\omega^2 \cos \theta}{h} \left[\frac{Rh^2}{2} - \frac{h^3 \sin \theta}{3} \right] = m\omega^2 h \cos \theta \left(\frac{R}{2} - \frac{h \sin \theta}{3} \right) \end{aligned}$$

Gravity torque about P , at the cm, $h/2$ up the rod:

$$\tau_g = -mg \frac{h}{2} \sin \theta$$

Balance. $\tau_c + \tau_g = 0$:

$$\omega^2 h \cos \theta \left(\frac{R}{2} - \frac{h \sin \theta}{3} \right) = g \frac{h}{2} \sin \theta \quad \implies \quad \omega^2 = \frac{g \tan \theta}{R - \frac{2h}{3} \sin \theta}$$

$$\omega = \sqrt{\frac{g}{R} \tan \theta \left(1 - \frac{2h}{3R} \sin \theta \right)^{-1}}$$

Checks: $h \rightarrow 0$ reduces to (a) ✓ Denominator $< R$, so ω is *larger* than in (a) ✓ — correct, since the rod's mass sits at smaller radius than R and so needs a faster spin to hold the lean.

↑ Two lines. The first proves the correction is consistent with (a); the second shows the sign of the correction was understood rather than stumbled into.

How that gets graded

USAPhO rubrics are not published, so the breakdown below is a reconstruction — but it is the right *shape*: a 25-point problem is cut into five or six milestones, and each is awarded on its own.

Rubric line	Pts	Earned
(a) Point-mass limit		
Correct force picture: states the frame and identifies the centrifugal force acting at the cm	2	2
Correct torque (or force) balance about the contact point	3	3
Correct $\omega = \sqrt{g \tan \theta / R}$	3	3
(b) Extended rod		
Recognizes the centripetal acceleration varies along the rod, so the cm substitution fails	3	3
Correct parametrization $r = R - s \sin \theta$, $z = s \cos \theta$, $dm = (m/h)ds$	4	4
Correct integral expression for the centrifugal torque	4	4
Correct evaluation $m\omega^2 h \cos \theta \left(\frac{R}{2} - \frac{h \sin \theta}{3} \right)$	3	3
Correct gravitational torque and balance	3	3
Correct final ω , boxed	3	3
Total	25	25

That submission gets full marks, and it is worth noticing how little of the page is polished. There is a crossed-out formula, an abandoned idea, and no prose beyond a handful of six-word fragments. **What makes it a 25 is that every rubric line is findable in under ten seconds.**

The same physics, written four other ways

Here is what actually separates scores on this exam. All four of these students did roughly the same amount of physics.

What the page contains	Score
Ran out of time in (b). Part (a) complete. In (b): “the centripetal acceleration varies along the rod, so I have to integrate the torque instead of using the cm,” plus $r = R - s \sin \theta$, $z = s \cos \theta$, $dm = (m/h)ds$, and the integral written down but never evaluated.	18/25
Fell into the trap. Part (a) complete. In (b), substituted $R \rightarrow R - \frac{h}{2} \sin \theta$ with no comment and boxed it.	9/25

What the page contains	Score
Blew part (a), recovered. Part (a) has the lever arms swapped and is never checked, so $\cot \theta$ is boxed. Part (b) is done correctly from scratch — correct setup, integral, evaluation, balance, and answer.	22/25
Did all the physics, wrote none of it. Everything correct, but done on scratch paper, with only the two boxed answers copied onto the answer sheets.	6/25

Read those four rows again. The third student got *part (a) wrong* and still scored 22, because subparts are graded independently and (b) was written out. The fourth student solved the entire problem correctly and scored 6, because the work was on the wrong paper.

The gap between 6 and 22 on this problem is not physics. It is where you wrote and how you labeled it.

Things that went wrong, and what to do about them

You suspect part (a) is wrong but cannot see why.

Run the limiting cases — that is what caught the swapped lever arms above. Set each parameter to zero or infinity in turn and ask whether the answer still makes sense. If a check fails, you have found a real error in thirty seconds. If they all pass, stop worrying and move on; a wrong (a) does not sink (b) anyway.

You cannot see how to set up the integral in (b).

Write down what you *do* know — “the acceleration varies along the rod, so $\tau_c = \int \omega^2 r z dm$ ” — and then define r , z , and dm one at a time against your diagram. That decomposition is worth 7 of the 17 points in (b) on its own, before you have integrated anything. Half of getting stuck is trying to write the whole setup in one line.

You get an ugly answer and think you must have erred.

Ugly is normal. $\omega^2 = g \tan \theta / (R - \frac{2h}{3} \sin \theta)$ is ugly, and correct. Check limits, box it, and move on. Do not spend ten minutes hunting for a mistake that isn’t there because you expected something prettier.

You have two versions of an answer and no time to decide.

Pick the one that passes your limiting checks, box that one, and put a single line through the other. Never leave both live.

You are 30 minutes in with (b) half done.

Stop. Box whatever you have, write one line saying what remains (“still need to evaluate the integral and balance against gravity”), and go read the next problem. Its first subpart is almost certainly worth more than your next ten minutes on this one.

7 Part VI: Two Real Papers

Everything so far has been a constructed example. These two are not. Both pages below are real 2018 USAPhO submissions to Problem B2, reproduced as the graders received them, with the scores those papers actually earned. The scans carry only the AAPT numbers printed on the answer sheet; no names appear on them.

Question B2 (2018 USAPhO). In this problem, use a particle-like model of photons: they propagate in straight lines and obey the law of reflection, but are subject to the quantum uncertainty principle. You may use small-angle approximations throughout.

A photon with wavelength λ has traveled from a distant star to a telescope mirror, which has a circular cross-section with radius R and a focal length $f \gg R$. The path of the photon is nearly aligned to the axis of the mirror, but has some slight uncertainty $\Delta\theta$. All small cross-sectional areas of the mirror are equally likely to include the point of reflection.

- (a) Find the standard deviation σ_r of the distribution for r , the distance from the center of the mirror to the point of reflection.
- (b) Use the uncertainty principle, $\sigma_r \sigma_{p_r} \geq \hbar/2$, to place a bound on how accurately we can know the angle of the photon from the axis of the telescope. Give your answer in terms of R and λ . *If you were unable to solve part a, you may also give your answer in terms of σ_r .*
- (c) Suppose we want to tell with high probability whether a detected photon came from the left half or right half of Alpha Centauri A. Approximately how large would the telescope have to be? The star is about 4×10^{16} m away with radius about 7×10^8 m; assume $\lambda = 500$ nm.

Read the italicized sentence in part (b) again. **The exam is handing you the escape hatch from Part II in writing.** Watch which of these two students takes it.

Paper 1 — 22 out of 25

(a) Variance $= \overline{r^2} = \frac{\int_0^R 2\pi r \cdot r^2 dr}{\pi R^2} = \frac{1}{2} R^2$. #4717
#69012
 $\hookrightarrow \Delta r = \frac{R}{\sqrt{2}}$. B2 - 1/1

(b)  B2 - 1/1
 $p = \frac{h}{\lambda} \rightarrow \Delta p_r = p \Delta \theta = \frac{h}{\lambda} \Delta \theta$
 $\hookrightarrow \frac{R}{\sqrt{2}} \cdot \frac{h}{\lambda} \Delta \theta \geq \frac{h}{2}$
 $\Delta \theta \geq \frac{1}{\sqrt{2}} \cdot \frac{\lambda}{R}$

(c) We want $\Delta \theta \leq \frac{7 \cdot 10^6}{4 \cdot 10^{16}} = 1.75 \cdot 10^{-8}$
 Rearranging above,
 $R \geq \frac{1}{\sqrt{2}} \cdot \frac{\lambda}{\Delta \theta}$
 $= \frac{1}{\sqrt{2}} \cdot \frac{5 \cdot 10^{-7}}{1.75 \cdot 10^{-8}}$
 $\approx 20 \text{ m.}$

What they did. All of B2 on one side of one sheet, headed B2 - 1/1. In (a) they wrote the area-weighted average straight down, $\int_0^R 2\pi r \cdot r^2 dr / \pi R^2 = \frac{1}{2} R^2$, and took the square root. In (b) they sketched the photon and the angle, wrote $p = h/\lambda$ and $\Delta p_r = p \Delta \theta$, substituted into the uncertainty relation, and boxed $\Delta \theta \geq \frac{1}{\sqrt{2}} \lambda/R$. In (c) they got the angle the telescope must resolve, $7 \times 10^8 / 4 \times 10^{16} = 1.75 \times 10^{-8}$, rearranged their own (b) for R , and boxed $\approx 20 \text{ m.}$

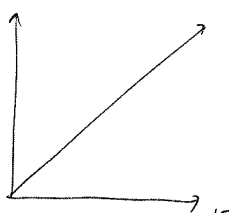
Why they got the credit they did. Each subpart is a short chain a grader can follow end to end in a few seconds, and both final answers are boxed. Note that **part (a) is actually wrong**: they computed the root-mean-square radius, not the standard deviation. With $\langle r \rangle = 2R/3$, the variance is $\langle r^2 \rangle - \langle r \rangle^2 = R^2/18$, so $\sigma_r = R/(3\sqrt{2})$ — smaller than what they wrote by a factor of three. They lost the points for (a) and kept essentially everything else, because (b) and (c) were carried forward correctly *from their own* (a). This is normal: graders mark consequentially, and a later part done right on an earlier number that is wrong still earns the later points.

What would have earned more. Only that one distinction. Writing down the definition they

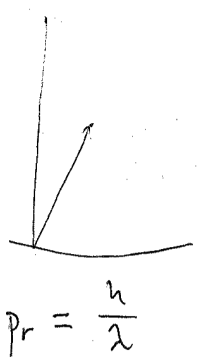
were using $\sigma^2 = \langle r^2 \rangle - \langle r \rangle^2$ — before computing anything would very likely have caught it, because the missing $\langle r \rangle^2$ term is visible the moment the formula is on the page. Naming the principle first is not only a courtesy to the grader; it is how you catch your own errors.

Paper 2 — 5 out of 25

a)



b)



c)

$$\Delta\theta < \tan^{-1}\left(\frac{7}{4} \times 10^{-8}\right)$$

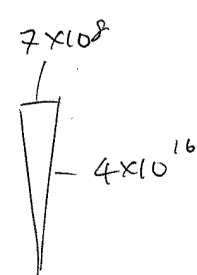
$$\Delta\theta < 1.75 \times 10^{-8}$$

Student # 4900

Proctor # 47982

Problem B2

Page 1 of 1



Before reading this one as a weak paper: the same student scored **25/25 on A1** and **23/25 on A3** that day, on a problem where 45% of the field scored zero. This is not someone who was out of their depth.

What they did. In (a), a pair of axes labeled r with an arrow across them — no distribution, no integral, no answer. In (b), a sketch of a photon striking the mirror and a single equation, $p_r = h/\lambda$. In (c), a sketch of the star subtending its angle at Earth, and $\Delta\theta < \tan^{-1}\left(\frac{7}{4} \times 10^{-8}\right) = 1.75 \times 10^{-8}$.

Why they got the credit they did. Two of the three physical ingredients on that page are right, and they are the same two Paper 1 used: the photon momentum h/λ , and the angular size of Alpha Centauri A. That is what the 5 points are for. But **nothing on the page connects them**. Part (b) never invokes the uncertainty principle it was asked about, so no bound on $\Delta\theta$ ever appears. Part (c) works out the angle the telescope must resolve and stops — the question asked how big the telescope must be, and no size is ever computed. Nothing is boxed.

What would have earned more. Three lines, none of which required an idea this student did not already have on the page:

- **Take the offer in part (b).** The problem says in so many words that you may answer in terms of σ_r . Writing $\sigma_r \cdot \frac{h}{\lambda} \Delta\theta \geq \hbar/2$, and solving for $\Delta\theta$, needs nothing whatsoever from part (a) — and $p = h/\lambda$ was already written down two inches above.
- **Finish (c) with the letters you have.** Even with no bound from (b), $R \sim \lambda/\Delta\theta = 5 \times 10^{-7}/1.75 \times 10^{-8} \approx 30$ m is one division, and it answers the question that was actually asked. An order-of-magnitude answer scores; a correct intermediate quantity that stops short of the question does not score much.
- **Give the sketches a caption.** Three of the marks on this page are drawings with no sentence attached. The (c) sketch encodes exactly the right idea — star radius over distance is the angle to be resolved — but a grader can only credit reasoning they can identify, and five words would have made it reasoning instead of a doodle.

Both students wrote the same two key facts on the page. One turned them into two boxed numbers and scored 22; the other left them as unconnected marks and scored 5. Neither outcome was about knowing more physics.

And this is not a story about one unlucky paper: B2's mean score was 5.0 out of 25, and **52% of the field scored zero on it**. Paper 2 is a completely typical B2 submission. The exam had told every one of those students, in the text of part (b), how to get past the step that stopped them.

8 Your Checklist

Before the exam

- ☐ Confirmed the date, start time, and site with your proctor
- ☐ Non-graphing scientific calculator, memory cleared
- ☐ Memorized the formulas the reference sheet does *not* give you — it is constants and four approximations, nothing else
- ☐ Practiced on at least two past USAPhOs *under timed conditions*, writing on paper, in 90-minute blocks

During the exam

- ☐ Read all three problems before starting any of them
- ☐ Diagram first, symbols defined on it
- ☐ Name the principle before the algebra
- ☐ Symbols throughout; numbers only at the end
- ☐ Box every subpart's final answer

- ☐ Header on every page: problem number and n/N
- ☐ Solve on the answer sheets, not on scratch
- ☐ Stuck? State an assumed result and do the next part anyway
- ☐ 30 minutes per problem, hard stop
- ☐ Last four minutes: box, label, cross out, sort

Questions this handout does not answer? Ask me — and check aapt.org/physicsteam, which is always the final authority.